

8 Forensic entomology is a method for estimating the time of death of a mammal.

- (a) Seventy-two hours or more after death, forensic entomology is the most accurate method for estimating the time of death.

Forensic entomology can also provide information about the place of death and indicate if the body has been moved.

- (i) Explain why forensic entomology is the most accurate method for estimating the time of death, if this is greater than 72 hours.

(2)

- (ii) Explain how forensic entomology can indicate if a body has been moved from the place of death.

(2)

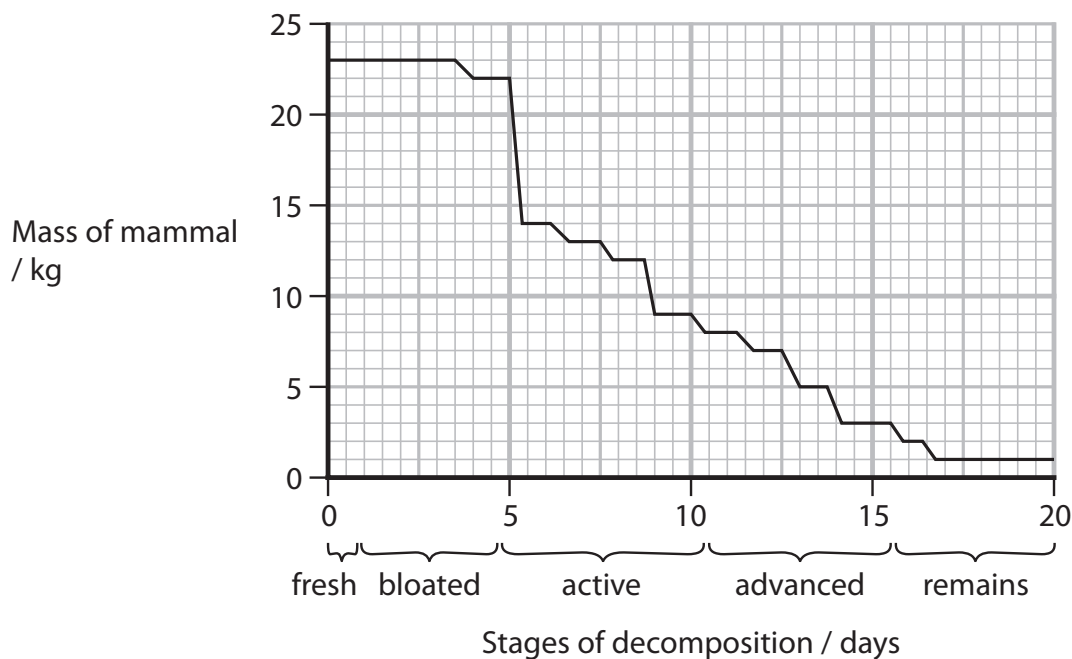
- (b) A study looked at the succession of insects associated with the decomposition of a dead mammal in the Andean Coffee region.

In this study, a mammal was killed, placed inside a metal cage in this region and left until it had completely decomposed.

The body of this mammal was monitored regularly.

The stages of decomposition were identified and various measurements were taken and recorded. Insects in the different stages of their lifecycle were collected and identified.

The graph shows the changes in mass of this mammal during the decomposition period.



- (i) Suggest why the mammal was placed inside a metal cage to decompose.

(1)

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- (ii) Calculate the rate of change in mass between day 5 and day 15 of decomposition.

Express your answer in kg hr^{-1} .

Give your answer to two decimal places.

(2)

Answer kg hr^{-1}

- (iii) Explain the changes in mass of this mammal during decomposition.

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- (iv) Suggest why some of the insect eggs, collected from the decomposing mammal, were taken back to the laboratory and kept for a few days.

(1)

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(v) The table shows data on some of the insects collected from this decomposing mammal.

Type of insect	Percentage of some of the insect found at each stage of decomposition (%)				
	Fresh	Bloated	Active	Advanced	Remains
<i>Lucilia</i>	100.0	61.4	1.8	3.7	0.0
<i>Cochliomyia</i>	0.0	15.5	68.4	35.1	0.0
<i>Chrysomya</i>	0.0	6.4	9.8	1.2	0.0
<i>Ophyra</i>	0.0	0.0	0.6	30.9	83.9
<i>Fannia</i>	0.0	0.0	0.0	0.1	0.0
Other types of insects	0.0	16.7	19.4	29.0	16.1

Explain how these results illustrate succession.

Use the information in the table to support your answer.

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(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 90 MARKS